Supratik Sarkar

Experimental Physicist & Photonic Engineer · Joint Quantum Institute, University of Maryland

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(Last updated in May 2026)

SUMMARY

Experimental physicist and photonic engineer with 6+ years of expertise in textbfphotonic integrated circuits, **quantum optics**, and nanofabrication across diverse material platforms including **silicon nitride**, **lithium niobate/tantalate**, **III-V semiconductors**, and **2D materials**. Proven track record of conceptualizing, designing, and characterizing nanophotonic devices and probing strongly correlated systems using cryogenic optical setups. Holds 3 provisional US patents. NSF I-Corps alumnus.

EDUCATION

University of Maryland (Joint Quantum Institute) **2021 – 2026**
Ph.D., Electrical and Computer Engineering (GPA: 3.9/4.0) *College Park, MD, USA*
Thesis: *Intersection of many-body physics and photonics: 2D semiconductors and photonic integrated circuits*
Advisor: Prof. Mohammad Hafezi

University of Waterloo (Institute for Quantum Computing) **2019 – 2021**
M.A.Sc., Electrical and Computer Engineering – Quantum Information (GPA: 96.2/100) *Waterloo, ON, Canada*
Thesis: *Creating and shaping light at single photon level* · Advisor: Prof. Michal Bajcsy

Jadavpur University **2014 – 2018**
B.E., Electronics and Telecommunication Engineering (GPA: 9.26/10, First Class with Honours) *Kolkata, WB, India*

RESEARCH EXPERIENCE

Ph.D. Researcher — Hafezi Group, Joint Quantum Institute **2021 – 2026**
University of Maryland *College Park, MD, USA*

- Designed **topological silicon nitride photonic circuits** (IQH/AQH lattices) generating nested frequency combs across 4 harmonic bands with 100% wafer-scale yield and relaxed phase-matching (*Science* 2025).
- Designed and experimentally demonstrated a **sub-wavelength 2D nanocavity** using atomically thin TMD mirrors; demonstrated chirality under magnetic field and electrical tunability (*Science Advances* 2024).
- Explored **Bose-Fermi-Hubbard physics** in WS_2/WSe_2 moiré heterobilayers by independently tuning fermionic and bosonic populations; observed excitonic Mott insulator state (*Science* 2026, *Nature Communications* 2024, *Physical Review X* 2024).
- Developed a **nonlocal sub-diffraction pump-probe scheme** using metasurface plasmon polaritons on TMD monolayers, achieving ∼2 orders of magnitude power reduction vs. free-space optics (*Science Advances* 2025).
- Designed PICs in **silicon nitride and lithium niobate** using ANSYS Lumerical, COMSOL, Zemax, and Tidy3D; performed tape-out and full optical characterization.
- Built and maintained **cryogenic optical setups** (Attodry, Montana 4K; Bluefors dilution fridge) for real-/momentum-space imaging, ultrafast spectroscopy, nonlinear optics, and time-tagged measurements.

M.A.Sc. Researcher — Bajcsy Group, Institute for Quantum Computing **2019 – 2021**
University of Waterloo *Waterloo, ON, Canada*

- Proposed deterministic **single-photon subtraction** from arbitrary states using quantum emitters coupled to chiral waveguides and cavities; demonstrated applications in generating non-classical states of light for quantum metrology.
- Designed diamond nanophotonic antennas via **adjoint optimization/inverse design** for directional emission from color centers; built confocal microscopy setup for characterization (*Science Advances* 2026).

Research Intern — Multimode Quantum Optics Group **Summer 2017**
Laboratoire Kastler Brossel (CNRS/ENS/Sorbonne) *Paris, France*

- Studied single-photon subtraction from continuous-variable cluster states (CNRS scholarship, *Physical Review Letters*).

SELECTED PUBLICATIONS & PREPRINTS

Full list: Google Scholar

- Single-shot realization of 10000-mode octave-spanning artificial gauge fields. *Under review in **Science*** (2026).
- Photon subtraction using a three-level emitter coupled to a chiral waveguide. *Under review in **npj Quantum Information*** (2026).
- Probing individual quantum emitters in bulk semiconductors via photonic nanojets. ***Science Advances*** 12, eaea5936 (2026).
  Media: IQC News; *Featured on Science Advances cover.*
- Probing quantum anomalous Hall states in twisted bilayer WSe_2 via attractive polaron spectroscopy. ***Physical Review***

X 16, 021037 (2026).

- Giant enhancement of exciton diffusion near an electronic Mott insulator. ***Science*** 391, 394 (2026).
  Media: JQI News, Phys.org
- Multi-timescale frequency-phase matching for high-yield nonlinear photonics. ***Science*** 390, 612 (2025).
  Media: JQI News, Phys.org, Optics.org, MIT Technology Review China
- Sub-wavelength optical lattice in 2D materials. ***Science Advances*** 11, eadv2023 (2025).
  Media: JQI News; *Featured on Science Advances website.*
- Chiral flat-band optical cavity with atomically thin mirrors. ***Science Advances*** 10, eadr5904 (2024).
  Featured on Science Advances website.
- Excitonic Mott insulator in a Bose-Fermi-Hubbard system of moiré WS_2/WSe_2 heterobilayer. ***Nature Communications*** 15, 2305 (2024).
- Tailoring Non-Gaussian Continuous-Variable Graph States. ***Physical Review Letters*** 121, 220501 (2018).

PATENTS

- **S. Sarkar**, M. J. Mehrabad, Y. Zhou, M. Hafezi. *Devices, systems, and methods for controllable, sub-wavelength light-matter interaction.* US Patent (Filed 2026).
- M. J. Mehrabad, L. Xu, **S. Sarkar**, Z.-Y. Wei, M. Hafezi. *Topological photonics architectures for optical computing and AI.* US Provisional Patent (Filed 2026).
- M. J. Mehrabad, L. Xu, A. Padhye, **S. Sarkar**, M. Hafezi. *Systems of ultrabroadband multimodal artificial gauge fields.* US Provisional Patent (Filed 2026).

TECHNICAL SKILLS

Photonics	Free-space & fiber optics, real-/Fourier-space imaging, ultrafast spectroscopy, interferometry, nonlinear optics, cryogenic optics (4K & mK), correlation measurements
PIC Design	Si_3N_4, $LiNbO_3$, III-V semiconductors; dispersion engineering, ring resonators, frequency comb design, topological lattices; ANSYS Lumerical, COMSOL, Zemax, Tidy3D, KLayout, GDSFactory
Nanofab	E-beam lithography, PVD, ALD, RIE, spin coating, reflectometry, wet bench
Programming	Python, MATLAB, Mathematica, C/C++, LabVIEW, Qiskit, TensorFlow, LaTeX
Equipments	Attodry & Montana cryostats, Bluefors dilution refrigerator, lock-in amplifiers, time-tagging electronics

AWARDS & HONORS

- Invention of the Year Award (Quantum category), University of Maryland (2026); Media: ECE News, JQI News
- NSF National I-Corps, National Science Foundation (NSF) (2026)
- ECE Distinguished Dissertation Fellowship, University of Maryland (2026)
- Outstanding Graduate Assistant Award, University of Maryland (2026); Media: JQI News
- Nanophotonics Poster Award, European Physical Society (EPS) NANOMETA (2026)
- Dean's Fellowship, University of Maryland (2021)
- WIN Nanofellowship, Waterloo Institute for Nanotechnology (2020)
- Faculty of Engineering Award (academic & research excellence), University of Waterloo (2020)
- Charpak Masters Scholarship, Govt. of France (2018, declined)
- CNRS Internship Scholarship, Laboratoire Kastler Brossel (2017)

SELECTED TALKS & CONFERENCES

- **Invited Talk**: META, Spain (2025)
- **Contributed Talk**: CLEO, USA (2025); PEPS, Iceland (2024); APS DAMOP (2021)
- **Breakthrough Talk**: NANOMETA, Austria (2024)

SERVICE & OUTREACH

Peer reviewer: *Physical Review B*, *Physical Review Letters*, *Physical Review Applied*, *Nano Letters*
Mentorship: Trained multiple PhD students in nanophotonic simulation and cryogenic experimental setups.
Scientific outreach: For the Joint Quantum Institute at Maryland Day.
Teaching: TA, ECE 404 Geometric and Physical Optics, University of Waterloo (Winter 2020)